MIL-M-21039B(AS)
11 June 1979
SUPERSEDING
MIL-M-21039A
25 MAY 1964

MILITARY SPECIFICATION

MAGNIFIER, MEASURING
TUBE TYPE, SEVEN POWER AR-60( )

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a seven-power precision magnifier with interchangeable reticles for interpreting and measuring photographic negatives, transparencies, and prints.

1.2 Classification. The magnifier shall be of one (1) type, designated AR-60( ). (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Air Engineering Center, ESSD (Code 93), Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

FSC 6650

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-155	- Joint Photographic Type Designation System.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Conflict with referenced documents. Where requirements stated in this specification conflict with any requirements in a referenced specification, the requirements of this specification shall apply.

3. REQUIREMENTS

3.1 First article. When specified by the procuring activity (see 6.2), the contractor shall furnish sample magnifiers for first article inspection and approval (see 4.5).

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be of good commercial quality entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect serviceability or appearance of the finished product.

3.2.1 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to atmospheric conditions likely to be encountered in storage or service use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

3.2.2 Screw threads. Screw threads used in the construction of the magnifier shall conform to the requirements of the FED-STD-H28.

3.3 Interchangeability. Similar units, parts, and accessories of each magnifier shall be directly and completely interchangeable with corresponding units, parts, and accessories of other magnifiers furnished on the same contract.

3.4 Components parts. The magnifier shall consist of the following component parts:

Lens
Lens mount
Supporting tube
Reticle

3.4.1 Lens. The lens shall be a precision ground, seven-power, and three element aplanat. Spherical and chromatic abberations shall be corrected to provide satisfactory coverage of a flat field having a minimum diameter of 25 millimeters.

3.4.1.1 Lens resolution. The lens shall be capable of resolving a minimum of 56 lines per millimeter at the center of the field and a minimum of 35 lines per millimeter at the edges of the field.

3.4.2 Lens mount. The lens shall be mounted in an eyepiece. Metal surfaces of the eyepiece shall be given a black anodic coating. The eyepiece shall be externally threaded to properly engage the threaded interior surface of the supporting tube to facilitate focus adjustment. Means shall be provided for maintaining an accurate fit between the threaded portion of the eyepiece and the threaded portion of the supporting tube.

3.4.3 Supporting tube. The supporting tube shall be a durable transparent plastic support capable of maintaining the lens in a fixed position at any selected distance from the image. The supporting tube shall be internally threaded to properly engage the threads of the eyepiece. The base of the supporting tube shall be externally threaded to properly engage the threads of the reticles. The supporting tube shall maintain the proper optical alignment of the lens and the reticle at all times.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

3.4.4 Reticle. The reticle shall be mounted in a metal cell. Glass used in the reticle shall be of good optical quality with no trace of color. The glass shall be free from striae, bubbles, seeds, stones, or other visible defects. The surfaces of the reticle shall be ground and polished parallel to each other within a maximum wedge angle of 5 minutes of arc. The reticle pattern shall be produced by the photoetch process. The lines produced by the photoetch process shall be filled with a cinnabar red filler. The shape, dimensions, and tolerances of reticles A through G shall be in accordance with Figures 1 through 7, respectively. The procuring activity shall specify the type and quantity of reticles to be supplied with the magnifier (see 6.2).

3.4.4.1 Reticle cell. Each glass reticle shall be permanently mounted in a metal cell. The metal cell shall be internally threaded to properly engage the exterior threaded surface of the base of the supporting tube. The metal cell shall hold the reticle pattern surface parallel to and not more than 0.3 millimeters from the image being viewed.

3.4.4.1.1 Reticle cell marking. The edge of each reticle cell shall be durably marked to indicate the type of reticle pattern and the manufacturer's name or trademark.

3.5 Case. Unless otherwise specified, a carrying case made of durable material shall be provided. The carrying case shall have a hinged, and tightfitting cover. The carrying case shall be sized to hold the magnifier, 5 reticles, and an instruction sheet. The magnifier and reticles shall be held in place in a manner to protect the glass surfaces from scratches, abrasions, and other damage.

3.6 Dimensions. The dimensions of the magnifier and carrying case shall be as specified herein.

3.6.1 Magnifier. The maximum dimensions of the magnifier shall not exceed 2.5 inches in length by 1.75 inches in diameter.

3.6.2 Carrying case. The maximum dimensions of the carrying case shall not exceed 5 inches in length, 2 inches in width, and 2 inches in height.

3.7 Weight. The maximum weight of the carrying case, with the magnifier and 5 reticles enclosed, shall not exceed 4 ounces.

3.8 Finish. The finish of the magnifier shall conform to the highest grade for precision optical instruments. The finish of the carrying case shall conform to the best commercial practice for high-grade drafting instrument cases.

3.9 Identification marking. The identification marking for the magnifier shall be placed on the exterior of the carrying case. The identification marking shall be in accordance with the requirements of MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

3.10 Workmanship. Workmanship shall be of the highest grade in accordance with the best commercial practice for precision optical instruments.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.5).
2. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. The inspection conditions shall be as specified herein.

4.3.1 Temperature and humidity. Unless otherwise specified, the magnifier shall be inspected under room ambient conditions of 73± 9 degrees F and relative humidity of 50± 10 percent.

4.4 Failure during inspections. Any characteristics exhibited by the magnifier during any inspection specified herein which would cause the magnifier to fail to meet any of the requirements of Section 3 of this specification, while in service, shall constitute failure of the magnifier to pass the inspection and may be the basis for rejection of the magnifier.

4.5 First article inspection. The inspection shall consist of all inspections specified herein.

4.5.1 First article sample. The quantity of first article samples shall be as specified by the procuring activity (see 6.2). The first article sample shall be representative of the production model magnifier.

4.6 Quality conformance inspection. This inspection shall consist of the following inspections:

a. Sampling inspections (see 4.6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

4.6.1 Sampling inspections. The sampling inspections shall be as specified herein. Sampling for these inspections shall be in accordance with MIL-STD-105 general inspection level II. The acceptable quality level (AQL) shall be as specified by the procuring activity (see 6.2).

4.6.1.1 Physical examination. Examine each magnifier visually and with gauges to determine conformance to specified requirements for materials, construction, fit and interchangeability of components, dimensions, weight, finish, marking, and workmanship. Determine that the externally threaded surfaces of the eyepiece properly engage the internally threaded surfaces of the supporting tube and permit focus adjustment. Determine that the externally threaded surfaces of the supporting tube properly engage the internally threaded surfaces of the reticles. Using suitable gauges measure the reticle scales to determine conformance to specified dimensions and tolerances. Visually determine that the glass used in the reticles shows no trace of color and is free from striae, bubbles, seeds, stones and other visible defects. Using suitable gauges determine that the glass surfaces of each specified reticle are parallel to each other within 5 minutes of arc. Using suitable gauges determine that the reticle cell holds the pattern surface of the reticle parallel to and not more than 0.3 millimeters from the surface of the image being viewed. Visually examine the carrying case to determine that the case is constructed in such a manner to properly hold the magnifier and 5 reticles and to protect all glass surfaces from scratches, abrasion, and other damage.

4.6.1.2 Magnification. Properly attach a reticle of known scale to the supporting tube of the magnifier. Using the lens of the magnifier, project the image of the reticle onto ground glass. Determine the size of the projected image. Divide the size of the projected image by the size of the image on the reticle and determine conformance to the requirements for a lens magnification factor of seven.

4.6.1.3 Distortion. Examine a suitable calibrated grid pattern through the lens of the magnifier. Determine that the grid pattern exhibits no evidence of distortion within the field of view of the lens.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

4.6.1.4 Chromatic Aberration. Using the magnifier, examine a suitable high contrast white test pattern on a black background. Determine that the test pattern is free from visually perceptible color fringes over the entire field of view of the magnifier.

4.6.1.5 Field of view. Using the magnifier, view a series of concentric circles having known diameters. Determine that the largest circle that is completely visible through the lens of the magnifier has a diameter of not less than 25 millimeters.

4.6.1.6 Resolution. Position a 1951 USAF, or equal, high contrast resolution test plate in the object plane of the magnifier at the center of the field of view. Adjust the magnifier lens for best focus. Determine that the magnifier resolves a minimum of 56 lines per millimeter. Repeat this procedure at randomly selected positions at the edge of the field of view and determine that the magnifier resolves a minimum of 35 lines per millimeter.

4.7 Rejection and retest. Rejected magnifiers shall be replaced or reworked to correct the defects, after which all necessary inspections shall be repeated. Rejected magnifiers shall not be resubmitted for inspection without furnishing complete particulars concerning previous rejections and the measures taken to correct the defects. When investigation indicates that the defects may exist in magnifiers previously accepted, complete particulars concerning the defects found, including recommendations for correction, shall be furnished to the procuring activity.

4.8 Destroyed specimen. All parts, specimens, or assemblies destroyed in making inspections required to determine compliance with this specification shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

4.9 Examination of packaging and packing. Examine the packaging and packing of the magnifier in accordance with the Quality Assurance Provisions of the specifications referenced in Section 5.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Level A packaging shall be as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

5.1.1.1 Unit packaging. The magnifier and reticles enclosed in the carrying case shall be packaged in accordance with Method 1A-8 of MIL-P-116 and over-boxed in a snug-fitting paperboard box conforming to PPP-B-676. Closure of the box shall be as specified in the appendix to PPP-B-676.

5.1.1.2 Intermediate packaging. Magnifiers, packaged as specified in 5.1.1.1, shall be placed in intermediate containers conforming to PPP-B-636. Intermediate containers shall be uniform in size and shape and quantities shall be of minimum tare and cube and shall contain multiples of four unit packages, not to exceed 12 unit packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 12 unit packages.

5.1.2 Commercial. Magnifiers shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A. Magnifiers packaged Level A as specified herein shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant and style optional. Closure of the containers shall be as specified in the appendix to PPP-B-636.

5.2.2 Commmercial. Magnifiers shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. Each unit package, intermediate container, and exterior container shall be marked in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use. The magnifier is intended to be used to aid photographic interpreters in making detailed inspections and precise measurements of photographic images.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of magnifiers required.
(c) First article samples required (see 3.1.).
(d) Quantity and type of reticles required (see 3.4.4.).
(e) Acceptable quality level (AQL) required (see 4.6.2.).
(f) Level of preservation, packaging and packing required (see Section 5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

6.3 Type designator. The parenthesis in the type designator will be replaced by a suffix letter and a production issue number, in accordance with the provisions of MIL-STD-155. This complete type designator shall then be used on nameplates, shipping records, in instruction books, and wherever reference is made to the specific equipment defined by the designator. As soon as possible after award of the contract, the contractor should apply to the contracting officer for such information.

Custodians:

Navy - AS

Review Activities:

Navy - AS

User Activities:

Navy - AS

Preparing Activity:

Navy - AS

(Project No. 6650-N101)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

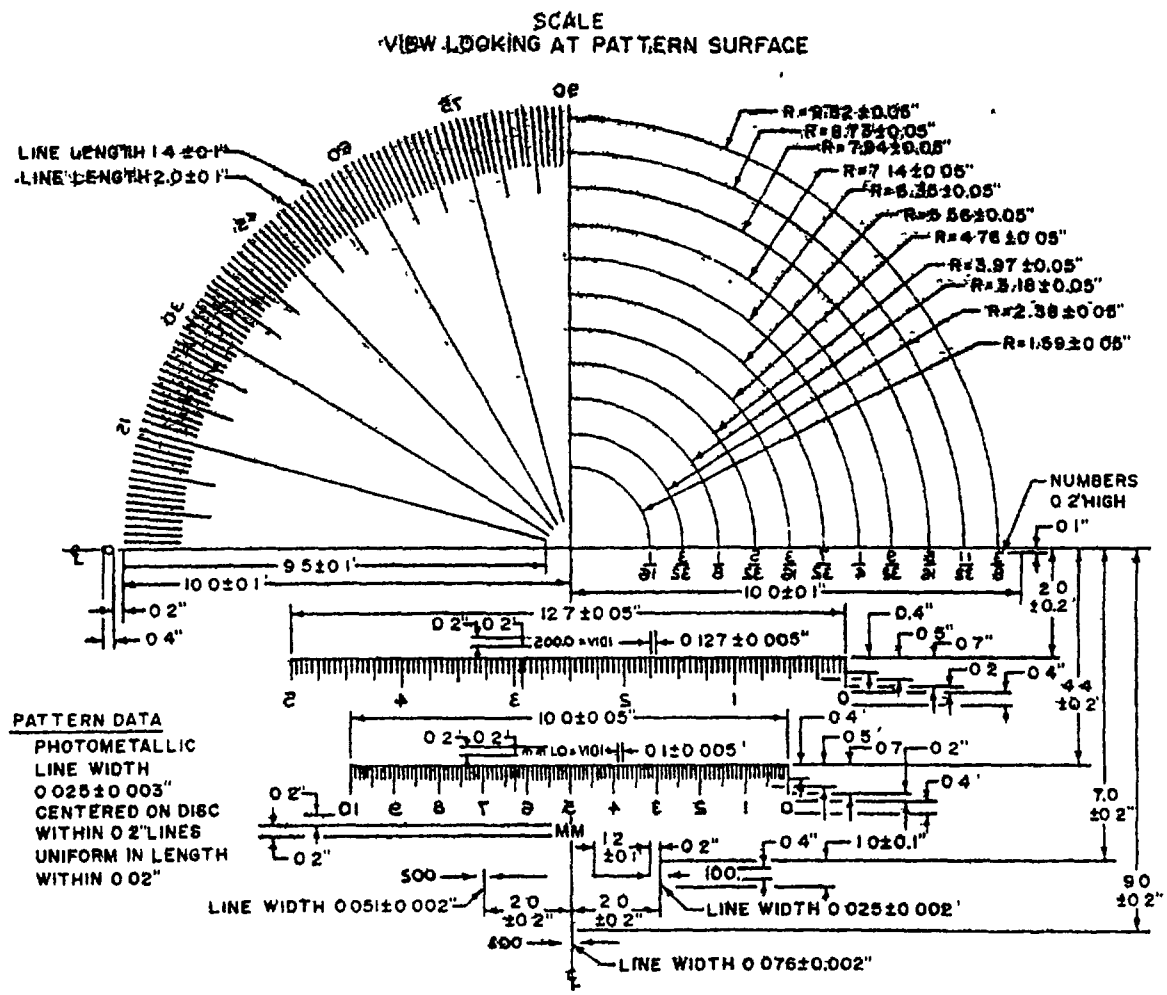
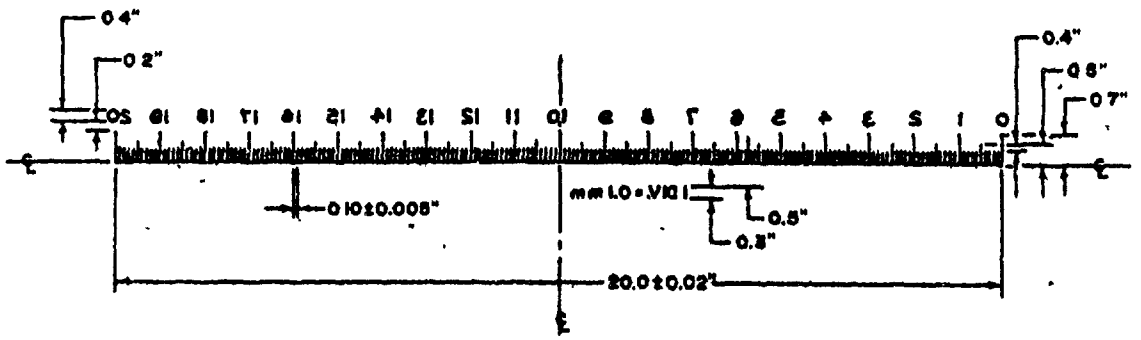


FIGURE 1 Reticle A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

SCALE METRIC

VIEW LOOKING AT PATTERN SURFACE



PATTERN DATA

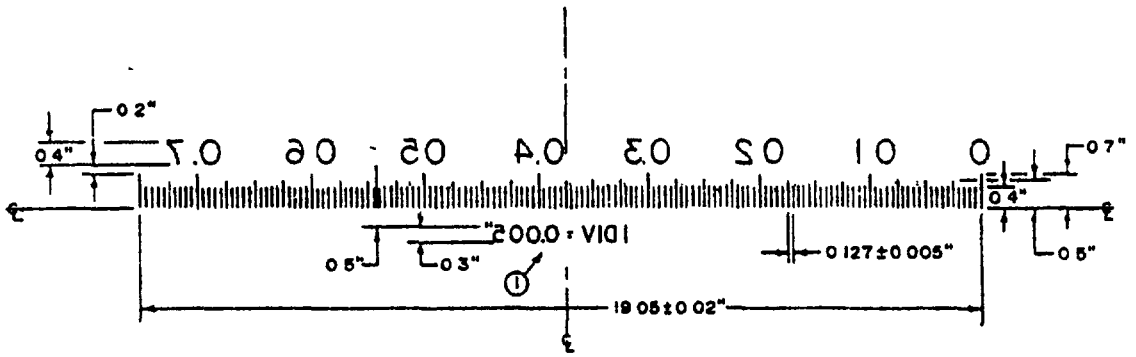
PHOTOMETALLIC LINE WIDTH 0.025±0.003"
LINES UNIFORM IN LENGTH WITHIN 0.02"
CENTERED ON DISC WITHIN 0.2"

FIGURE 2 Reticle B

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

SCALE - INCH

VIEW LOOKING AT PATTERN SURFACE



PATTERN DATA

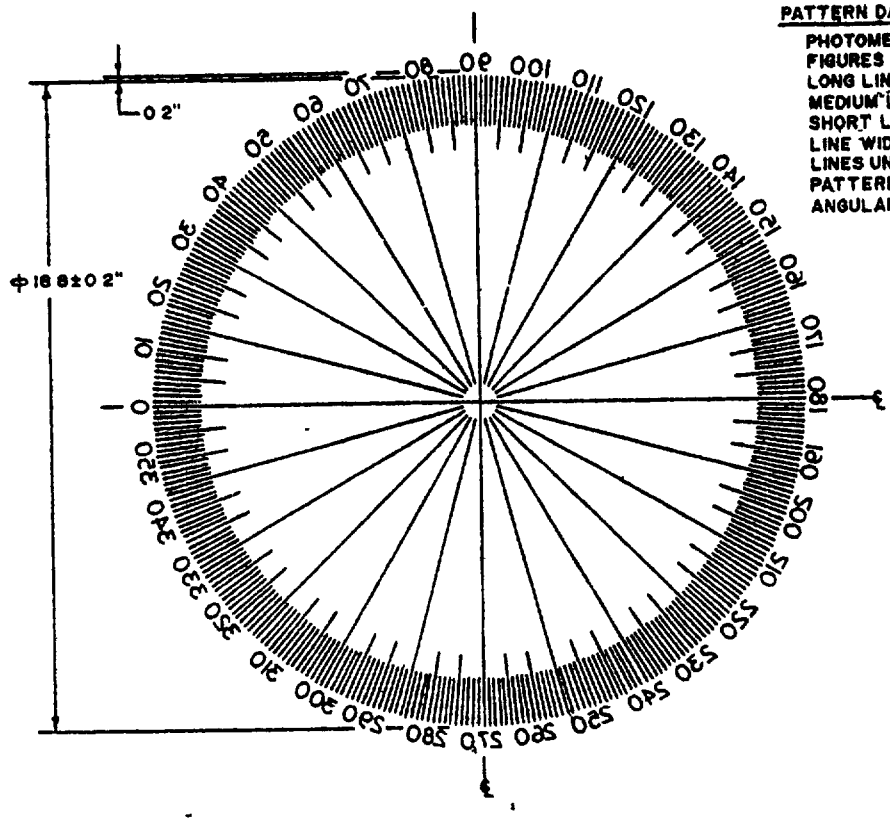
PHOTOMETALLIC LINE WIDTH 0.025±0.003"
LINES UNIFORM IN LENGTH WITHIN 0.02"
CENTERED ON DISC WITHIN 0.2"

FIGURE 3 Reticle C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

SCALE - PROTRACTOR

VIEW LOOKING AT PATTERN SURFACE



PATTERN DATA

PHOTOMETALLIC
FIGURES 0.4" HIGH
LONG LINES 0.90"
MEDIUM LINES 2.00"
SHORT LINES 1.55"
LINE WIDTH 0.025±0.003"
LINES UNIFORM IN LENGTH WITHIN 0.02"
PATTERN CENTERED ON DISC WITHIN 0.2"
ANGULAR TOLERANCE ±5'

FIGURE 4 Reticle D

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

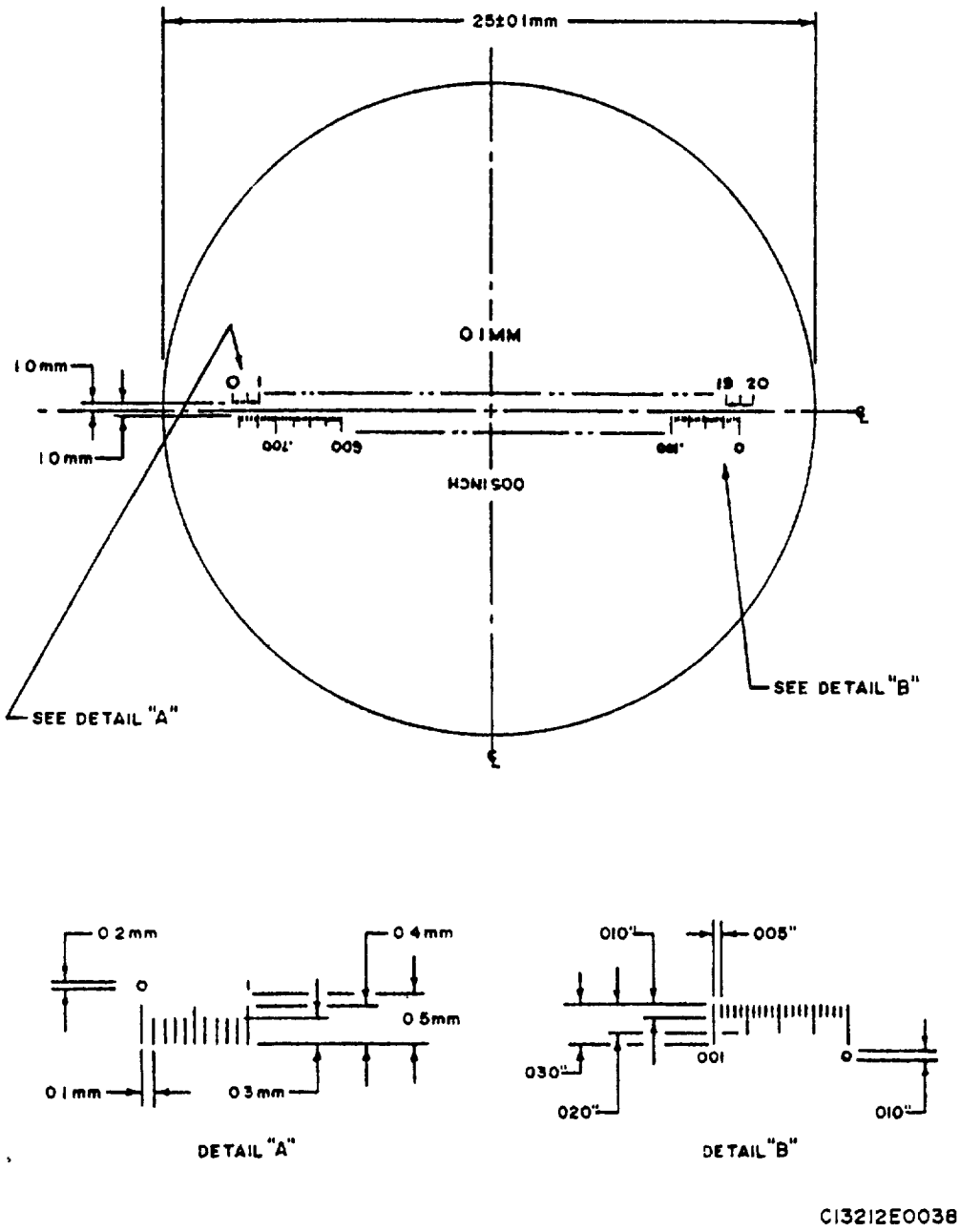
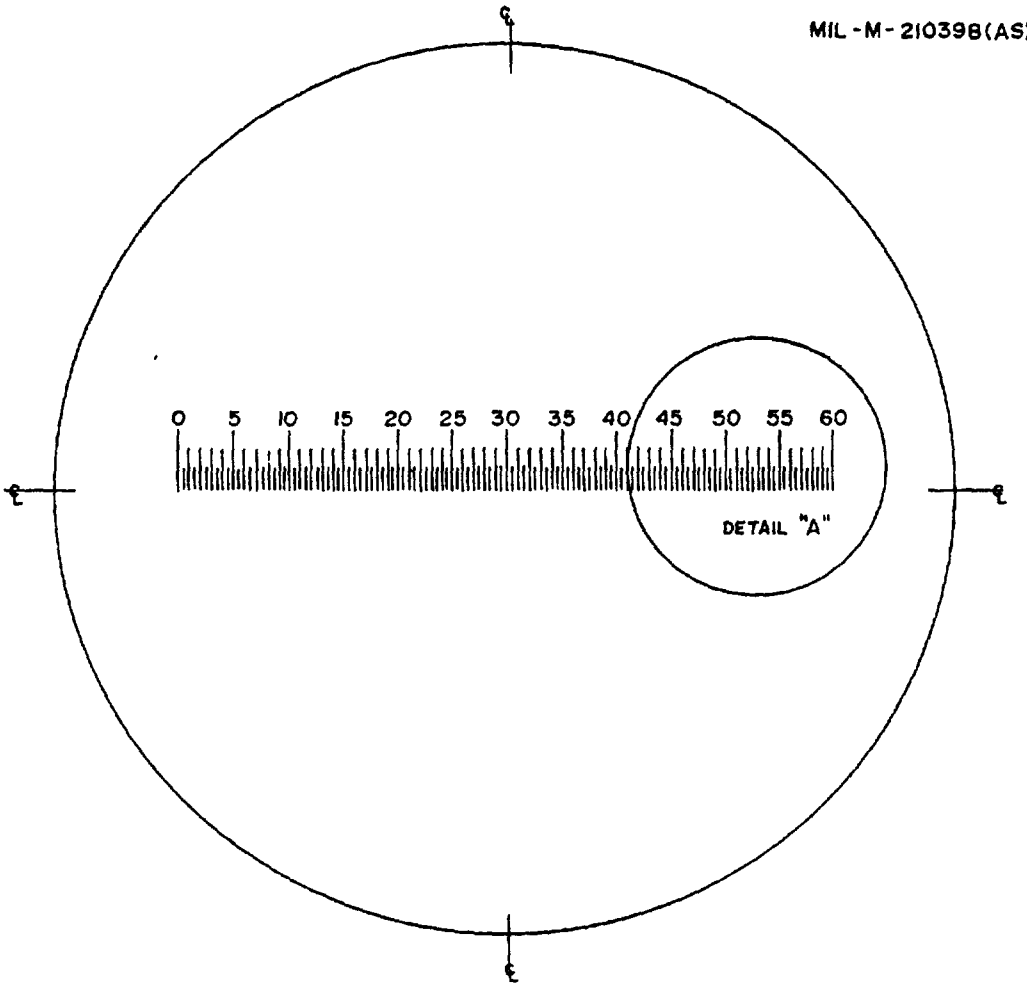
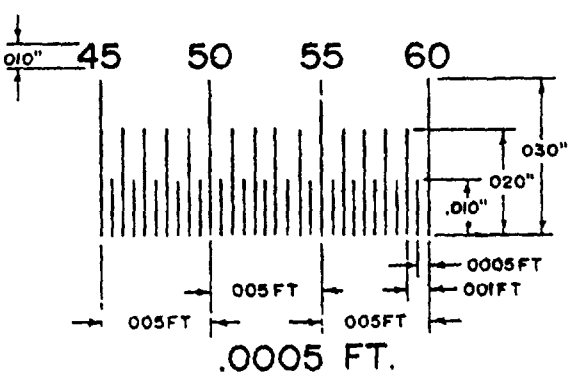


FIGURE 5 Reticle E

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.





DETAIL 'A"

FIGURE 6 Reticle F

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

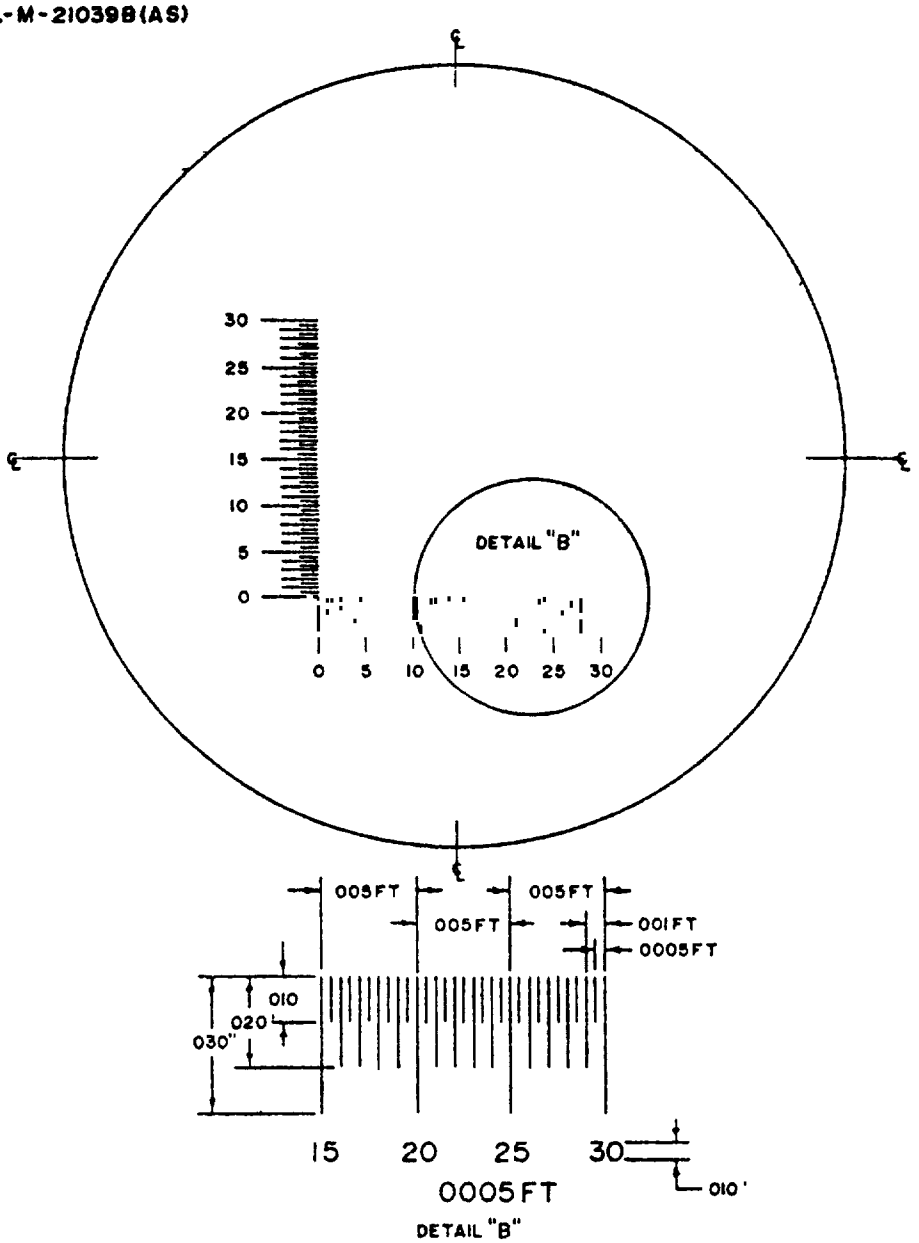


FIGURE 7 Reticle G

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-M-21039B(AS) Magnifier, Measuring Tube Type, Seven Power AR-60 ( )

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Naval Air Engineering Center
Engineering Specifications and Standards Department
(Code 93)
Lakehurst, N.J. 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:35Z
Check the source to verify that this is the current version before use.